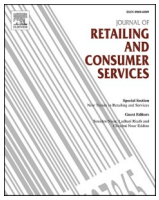
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The effects of regulatory focus, processing disfluency, and shopping goals on consumer perceptions of range-of-savings claims in retailers' mobile ads

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ABSTRACT

This research investigates three factors influencing consumer perceptions of range-of-savings claims in mobile ads. Since such claims are inherently ambiguous, consumers perceive uncertainty and make judgments about the discount through an anchor-adjustment process. We expect that consumers' regulatory focus will differentially affect these psychological responses. However, when savings claims are presented in a disfluent manner, this effect will vary depending on whether the consumer has a shopping goal. Specifically, consumers with a shopping goal engage in more deliberate and systematic processing, leading them to modify their initial responses. In contrast, consumers without a shopping goal are likely to rely on automatic and heuristic processing, reinforcing the influence of naïve theories about disfluency. To test the hypotheses, we conducted an experiment with three hundred seventy-two South Korean adults using a 2 x 2 x 2 between-subjects design. The results showed that when a shopping goal was present, promotion-focused participants perceived the discount as less credible, judged the product's value lower, and showed weaker purchase intentions when the savings claim was disfluent (vs. fluent). In contrast, prevention-focused participants' perceptions of credibility, product value, and purchase intentions were higher in the disfluent conditions. When no shopping goal was present, participants responded more positively to disfluent savings claims, regardless of their regulatory focus. This study extends our understanding of tensile price claims and regulatory focus, provides new insights into the mechanisms of processing fluency, and offers guidelines for designing effective pricing strategies and online ads, including customization.

1. Introduction

Price discounts have long been employed by retailers as a key strategy in marketing and promotion. Retailers often manage a wide range of products, including different categories or various models within a single category, and apply varying discount levels depending on the product. During such promotional events, although precise discount information is provided for individual products, the advertisements for these events—designed to attract customers to retail stores—frequently convey ambiguous discount information. For instance, retailers may present consumers with the entire range of discounts (e.g., 30%–70% sale) or specify only the minimum (e.g., starting at 30% off) or maximum savings (e.g., up to 70% savings), a strategy known as tensile price claims. These promotions are communicated via various channels, such as advertisements, websites, and in-store display. In mobile advertisements, due to space constraints, discount information typically becomes the primary focus.

Considering their importance and frequent use in practice, tensile price claims have received significant scholarly attention. Most studies have focused on comparing tensile price claims with precise discount claims or other types of discounts, or on evaluating the relative effects of the three forms of tensile price claims (Biswas and Burton, 1994; Mobley et al., 1988). Potential moderators, such as the level of the discount and the proportion of discounted items, have also been examined. However, to the best of the authors' knowledge, no research has focused on specific types of tensile price claims and explored the factors that determine their effectiveness. To address this gap, the present research focuses on "range of savings" claims. Unlike maximum or minimum savings claims, which present a single reference point, range-of-savings claims offer two distinct anchors that consumers use to evaluate the deal's value. This dual-anchor structure increases variability in interpretation, underscoring the need to understand the factors that shape how consumers

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perceive and respond to these claims. Therefore, we aim to examine the type of consumer, the processing difficulty of the claims, and the situational context in which they are encountered as key factors.

More specifically, the ambiguity and dual anchors associated with range-of-savings claims highlight the role that consumers' regulatory focus plays in determining their reactions to such claims. It is well-known that consumers respond differently to uncertainty and minimal-maximal goals, depending on their type of focus (Shah and Higgins, 1997). Additionally, consumers' perception of claims is likely influenced by how the information is presented and processed. This research focuses on the concept of processing fluency, specifically investigating how disfluent (vs. fluent) presentation affects consumers' experience about processing information and their subsequent responses to the claims. Furthermore, we propose that the impact of processing disfluency varies depending on the consumer's goal during the purchase decision-making process. Consumers encounter discount advertisements either while actively seeking purchase-related information (shopping goal present) or during routine activities without an intent to purchase (shopping goal absent). The presence or absence of a shopping goal influences consumers' cognitive system and information processing styles, interacting with processing disfluency in complex ways (Kahneman, 2003; Lee and Ariely, 2006).

The findings of this study contribute to a deeper understanding of tensile price claims, regulatory focus, and processing fluency and are expected to provide insights into other forms of range-based information (e.g., expected returns or probability estimates in surveys). They also offer valuable practical guidelines. Mobile advertisements, a widely used medium today, present opportunities for highly personalized and targeted marketing. Firms leveraging online consumer data can identify regulatory focus tendencies and shopping goals—key factors in this study—enabling them to create customized advertisements that vary in processing fluency based on the target audience.

The paper is structured as follows. A review of prior research on tensile price claims positions this study within the literature and provides the theoretical background. The subsequent sections explore how consumers' regulatory focus affects their reactions to range-of-savings claims and examine how this relationship shifts when the information is presented disfluently. The role of shopping goals in moderating these effects is then proposed. The study employs an experimental methodology, the design and results of which are discussed in detail. The paper concludes by summarizing key findings, discussing theoretical and practical implications, and outlining limitations and directions for future research.

2. Literature review

Tensile price claims refer to the ambiguous presentation of price information, typically applied when retailers offer varying discount or savings rates across different products. These claims generally fall into one of three categories: maximum-savings, minimum-savings, or range-of-savings claims (Biswas and Burton, 1993; Mobley et al., 1988). Mobley et al. (1988), considered a pioneer in tensile price claims research, compared the effects of maximum-savings claims to precise discount claims. They found that consumers were less likely to trust maximum-savings claims, perceived the overall value of such offers as lower, and estimated actual savings to be significantly less than advertised. Furthermore, these effects were more pronounced at higher discount levels. This suggests that the inherent ambiguity of maximum-savings claims increases consumers' perceptions of uncertainty, leading to skepticism and a lack of credibility. In a subsequent study, maximum-savings claims were compared with Scratch-and-Save promotions, where the actual discount is revealed only after purchase. Interestingly, consumers perceived these promotions as even more uncertain than tensile price claims, making maximum-savings claims more attractive (Choi et al., 2010).

Dhar et al. (1999) further examined range-of-savings claims in contrast to precise discount claims and discovered that the effects varied depending on the proportion of discounted items within the inventory. When the proportion of discounted items was high, consumers judged offers with precise claims as more attractive, consistent with earlier research. Conversely, when the proportion was low, range-of-savings claims elicited more positive reactions. This response could be attributed to experiences in situations like lotteries, where a negative correlation exists between the frequency of occurrence and the magnitude of the reward. Consumers inferred that a smaller proportion of discounted items indicated greater discount magnitudes (Dhar et al., 1999). Additionally, under precise discount claims conditions, consumer perceptions of discounts closely matched the advertised rates, whereas for range-of-savings claims, perceived discount levels fell below the advertised values, reinforcing earlier findings that consumers tend to undervalue tensile price claims (Mobley et al., 1988).

Scholars have also assessed the relative effects of different types of tensile price claims. According to Biswas and Burton (1994), consumers perceived savings and price reductions as larger and exhibited higher shopping intentions in the following order: minimum-savings, range-of-savings, and maximum-savings claims. Contrary to expectations, the moderating effect of store type was not significant. Furthermore, Biswas and Burton (1993) demonstrated that the relative effectiveness of the three claim types varied with the breadth of the discount range. Specifically, when discount ranges were broader, consumers perceived savings, the value of the deal, and the price reduction as larger and showed more favorable attitudes toward the deal for maximum-savings claims than for minimum-savings or range-of-savings claims. Conversely, when discount ranges were narrower, minimum-savings and range-of-savings claims elicited more positive attitudes and higher estimates of price reduction compared to maximum-savings claims. These differences were explained through the anchoring-adjustment mechanism (Tversky and Kahneman, 1974). Consumers use the specified discount levels as anchors and adjust their judgments accordingly. Since adjustments tend to remain closer to the anchor, downward adjustments from maximum-savings claims result in higher perceived value than upward adjustments from minimum-savings claims (Epley and Gilovich, 2005). More recently, Banerjee et al. (2016) compared just-below frames (e.g., 10%–39%) and round frames (e.g., 10%–40%) for range-of-savings claims. Consumers responded more positively to just-below frames in terms of expected average discounts, perceived savings, and overall deal value. This was attributed to the greater precision of just-below frames, which enhanced the credibility of the discount information and reduced the ambiguity and uncertainty often associated with tensile price claims.

In summary, prior research on tensile price claims has primarily focused on comparing their effects with other types of discounts or examining the relative effectiveness of different types of tensile price claims. Most of these studies have explored moderating factors related to discount- or retailer-specific characteristics. Additionally, research has indicated that the effectiveness of tensile price claims can vary depending on the type of outcome variables. Therefore, this study focuses on a specific type of tensile price claim, identifying previously underexplored consumer-related factors and examining various dimensions of consumer responses.

3. Conceptual development

3.1. Overview

According to the literature review, tensile price claims lead consumers to perceive ambiguity and uncertainty. In particular, range-of-savings claims allow for greater consumer interpretation due to the presence of two reference points—the minimum and maximum savings—within the anchor-adjustment mechanism used to infer the amount saved. Therefore, we first propose that consumers' regulatory focus plays a crucial role in this process. This expectation is based on

well-established findings that consumers with different types of regulatory focus exhibit varying sensitivities to uncertainty and risk, as well as different tendencies in goal setting and achievement (e.g., minimum vs. maximum goals, loss avoidance vs. gain pursuit).

Additionally, when consumers perceive and respond to tensile price claims, both the content and presentation format influence the processing of information. Drawing on the concept of processing fluency, this study predicts that when information is presented in a disfluent manner, the influence of consumers' regulatory focus will differ. Furthermore, processing disfluency is expected to exert distinct effects through different mechanisms, depending on consumers' cognitive systems and information processing styles. In this research context, we argue that when consumers encounter tensile price claims, their cognitive system and processing style will vary based on whether they have a shopping goal in mind. In conclusion, the interaction effect of regulatory focus and processing disfluency on tensile price claims is moderated by the presence or absence of a shopping goal. The research model is graphically depicted in Fig. 1, and the next section provides a detailed explanation.

3.2. Regulatory focus

Regulatory focus is a motivational system that governs how individuals regulate their behaviors to achieve goals. It is categorized into two distinct types: promotion focus and prevention focus (Crowe and Higgins, 1997; Higgins, 1997). Promotion focus is associated with aspirations, ideals, and hopes, encouraging individuals to pursue progress and achievement. These individuals are particularly attuned to positive outcomes, such as gains and opportunities. In contrast, prevention focus is closely related to obligations, duties, and responsibilities, emphasizing protection and safety. Prevention-focused individuals aim to avoid errors and losses, exhibiting heightened sensitivity to negative outcomes. Regulatory focus also shapes information processing styles. Prevention-focused consumers adopt a vigilant and conservative approach to avoid risks, prioritizing accuracy and careful evaluation, whereas promotion-focused consumers engage in exploratory and risk-taking information processing, driven by their focus on goal pursuit and achievement (Zhu and Meyers-Levy, 2007).

When applied to range-of-savings claims, the types of regulatory focus are expected to lead to different consumer reactions. Range-of-savings claims, which present only the minimum and maximum discount levels, create uncertainty about the actual discount for a specific product. This ambiguity reduces clarity for consumers. As a result, prevention-focused consumers, who prioritize risk avoidance and accuracy, are more likely to distrust such claims. In contrast, promotion-focused consumers, who seek opportunities and take risks, are less concerned with this uncertainty. Believing they can control the outcome, they are more likely to perceive range-of-savings claims as trustworthy opportunities (Langens, 2007).

In addition, consumers often activate persuasion knowledge in response to marketing activities such as price discounts (Kirmani and Zhu, 2007). Prevention-focused consumers, who engage in vigilant information processing, are more likely to infer hidden or ulterior motives from range-of-savings claims that provide incomplete information, leading them to perceive such claims as less credible (Xie and Kahle, 2014). In contrast, promotion-focused consumers are less likely to scrutinize such claims closely or interpret them as manipulative, as they tend to focus on potential opportunities rather than risks.

Furthermore, when evaluating range-of-savings claims, consumers are likely to consider both the potential satisfaction from obtaining a desired discount (outcome) and the likelihood of achieving it (probability of attainment) (Shah and Higgins, 1997). Promotion-focused consumers prioritize the outcome over the likelihood, often setting ambitious, maximal goals (Crowe and Higgins, 1997; Shah and Higgins, 1997). As a result, they tend to focus on the upper limit of the discount range, anchoring their expectations at the midpoint and adjusting upward toward the maximum discount (Biswas and Burton, 1993). Even when facing the risk of disappointment, their adventurous and risk-tolerant nature compels them to prefer the maximum savings, as it offers the highest potential reward. In contrast, prevention-focused consumers, who are inherently cautious and risk-averse, adjust their expectations downward toward the minimum savings. This minimizes the likelihood of negative deviations from their expectations and aligns with their preference for safety and certainty (Shah and Higgins, 1997). For these consumers, estimates closer to the minimum discount while offering smaller rewards, are perceived as safer alternatives with a higher level of certainty. Indeed, prevention-focused consumers, who emphasize the fulfillment of obligations or responsibilities, are known to set minimal goals and exhibit a preference for safer choices (Crowe and Higgins, 1997; Shah and Higgins, 1997).

In summary, when exposed to range-of-savings claims, prevention-focused consumers are more likely to perceive the claims as less credible and anticipate a lower average discount percentage compared to promotion-focused consumers. Regarding the two outcome variables, we propose that they are not only independently influenced but also interrelated. Specifically, perceived claim credibility can influence consumers estimates of potential savings (Carlson and Kukar-Kinney, 2018). If consumers find range-of-savings claims untrustworthy, they are more likely to lower their estimated savings or anchor to the minimum savings rather than the maximum as a precautionary measure. This suggests a mediating relationship between claim credibility and expected average savings. In turn, perceived average savings play a crucial role in determining consumers' perceptions of product value, ultimately influencing their intentions to visit and purchase from the retailer.

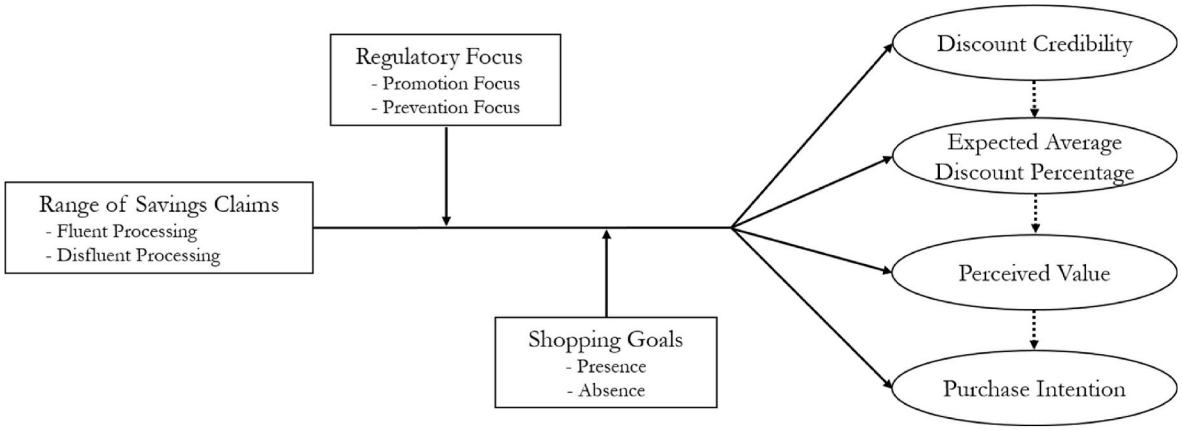


Fig. 1. Research model.

(Grewal et al., 1998). However, we argue that the effect of regulatory focus varies depending on additional factors, which will be discussed in the following sections.

3.3. Processing fluency

When retailers present range-of-savings claims through advertisements or other channels, the information can be delivered in various formats. Consumers, in turn, may perceive the processing of this information as either easy or difficult, a concept known as processing fluency (Schwarz, 2004). In particular, in online and mobile environments, large volumes of information are often presented in complex ways, incorporating various visual elements (Wu et al., 2016). This complexity makes it more difficult for consumers to process the information. While advertising is typically designed to enhance fluency, what happens when information is processed in a disfluent manner?

Research identifies three primary mechanisms through which processing fluency operates. First, when information is processed fluently, consumers experience positive affect, which is transferred to more favorable evaluations of the target (fluency-affect link; Forster et al., 2013; Reber et al., 1998). For instance, when consumers process an image fluently, they experience positive emotions, leading to higher evaluations of its aesthetic appeal and overall likability (Reber et al., 1998).

Second, consumers attempt to rationalize their experience of fluency or disfluency by activating naïve theories based on existing knowledge (Schwarz, 2004). These theories shape their interpretations and evaluations. For example, fluently processed information may be perceived as familiar, prompting the assumption that "this must be something I recognize or a common event" (Song and Schwarz, 2009). Conversely, disfluently processed information can trigger a metacognitive response, leading consumers to perceive it as novel or unique (Pocheptsova et al., 2010). Consequently, fluently processed advertisements for everyday consumer products elicit perceptions of familiarity and increase purchase intentions. However, for special-occasion products, where exclusivity is valued, disfluently processed advertisements evoke associations with uniqueness, resulting in stronger purchase intentions (Pocheptsova et al., 2010).

Finally, processing fluency influences consumers' information processing styles. When information is perceived as disfluent, it signals caution and prompts consumers to adopt more systematic and analytical processing strategies (Alter et al., 2007). This shift often results in adjustments from initial judgments and reduces reliance on heuristics and superficial cues, focusing on core attributes. For example, disfluent processing has been shown to enhance recall of discount details and foster preferences for such offers (Motyka et al., 2016). Additionally, in learning environments, disfluent processing encourages deeper engagement with complex tasks, leading to fewer errors (Song and Schwarz, 2008).

For range-of-savings claims, we argue that the second mechanism (naïve theories) and third mechanism (information processing style) are particularly relevant, as the processing of price information is inherently cognitive. Affect-based mechanisms are expected to play a minimal role. Prior research on processing fluency in the context of price discounts supports the theoretical validity of the cognitive mechanisms, though the specific topics addressed differ from those of the present study (Thomas and Morwitz, 2009). Moreover, the influence of these mechanisms is expected to vary depending on the context in which consumers encounter range-of-savings claims, as discussed in the following section.

3.4. Presence or absence of a shopping goal

Consumers may encounter tensile price claims either while actively searching for purchase-related information or evaluating alternatives (i.e., when a shopping goal is present) or while engaging in daily routines without a specific intent to purchase (i.e., when a shopping goal is absent). The presence or absence of a shopping goal influences the cognitive systems and information-processing styles consumers use (Lee and Ariely, 2006). Furthermore, this difference affects how consumers respond to the disfluency of such claims.

When consumers have a shopping goal, they are likely to be deliberate, controlled, and effortful in their judgment and decision-making (System 2 thinking), processing information carefully and deeply (systematic processing; Celsi and Olson, 1988; Chaiken, 1980; Kahneman, 2003). More specifically, they seek information aligned with their goals and are motivated to process relevant details, willingly allocating significant cognitive resources. Under such conditions, if consumers encounter range-of-savings claims presented in a disfluent manner, they are likely to exert greater effort and engage in more systematic and deliberate information processing and judgment. This deeper processing reduces reliance on pre-existing beliefs, thereby mitigating confirmation bias (Hernandez and Preston, 2013) and encourages judgments that deviate from initial impressions (Wegener and Petty, 1995).

As previously discussed, during fluent information processing, promotion-focused consumers—characterized by adventurous tendencies and the pursuit of maximal goals—generally respond positively to range-of-savings claims. Conversely, prevention-focused consumers, who are vigilant and prioritize achieving minimal goals, tend to react more negatively to such claims. However, under disfluent information processing, these response patterns are likely to shift. Disfluency prompts consumers to reassess their habitual tendencies.

Specifically, promotion-focused consumers' positive perceptions are expected to become less positive, while prevention-focused consumers' negative judgments are likely to become less negative. Promotion-focused consumers, who typically find range-of-savings claims credible and align their expectations with the upper bound of the discount, are likely to reassess the credibility of the claims and adjust their estimates toward the midpoint anchor. In contrast, prevention-focused consumers, who often approach such claims with skepticism and gravitate toward the lower bound, are expected to increase their trust and shift their estimates of discount percentage toward the midpoint anchor. This adjustment leads to a convergence in perceptions, reducing the disparity between the two groups. This leads to the following hypothesis:

H1a. When consumers have a shopping goal, promotion-focused consumers will (1) perceive the discount claims as less credible, (2) expect the average discount percentage to be lower, (3) judge the value of the product as lower, and (4) exhibit weaker intentions to purchase from the retailer when the processing of range-of-savings claims is disfluent (vs. fluent). In contrast, prevention-focused consumers will (1) perceive the discount as more credible, (2) expect the average discount percentage to be higher, (3) judge the value of the product as higher, and (4) exhibit stronger intentions to purchase from the retailer when the processing of range-of-savings claims is disfluent (vs. fluent).

On the other hand, when consumers do not have a shopping goal, they are more likely to adopt System 1 thinking and a heuristic processing style (Celsi and Olson, 1988; Chaiken, 1980; Kahneman, 2003). In such instances, they exhibit lower motivation to engage with savings advertisements, allocate fewer cognitive resources, and process information in a passive, heuristic manner (Niza Braga and Jacinto, 2022). Additionally, their attention becomes more stimulus-driven, focusing on superficial attributes of the stimuli (Bialkova and van Trijp, 2010), and they rely on automatic and immediate judgments.

Under these conditions, we expect that the disfluent processing of range-of-savings claims will likely activate naïve theories stored in consumers' memory, directly influencing their perceptions and judgments. Research has shown that the activation and influence of naïve theories involve automatic inferences, particularly when consumers have low motivation and limited cognitive resources for logical reasoning (Inman et al., 1997; Schwarz, 2004). In other words, the naïve theory mechanism underlying processing disfluency aligns with the use

of System 1 thinking and heuristic processing styles by consumers. But which specific naïve theories are likely to be activated?

Previous research suggests that fluently processed information is often interpreted as "familiar" (Song and Schwarz, 2009). Conversely, when the processing of ad copy or brand names is difficult, consumers are inclined to perceive the target as "unique or new" (Pocheptsova et al., 2010). Price discount advertisements are typically designed for fluent processing to ensure clear communication. In such cases, the experience of fluent processing leads consumers to activate a naïve theory of familiarity, prompting an inference that the promotion is common and takes place frequently. On the other hand, when range-of-savings claims are perceived as disfluent, consumers are likely to activate a naïve theory of unfamiliarity, inferring that the discount is rare or uncommon (Eisenbeiss et al., 2015). This inference has notable implications for consumer responses. Rare or uncommon discounts are perceived as more valuable and desirable due to the unavailability heuristic (Mukherjee and Lee, 2016). Moreover, consumers in these contexts lack the motivation or cognitive resources to engage in the anchor-adjustment process, which is the primary mechanism underlying the impact of tensile price claims. Consequently, the differentiated effects of regulatory focus based on these processes are expected to diminish, as these effects are typically more pronounced in a deliberative and systematic mode. Based on this reasoning, the following hypothesis is proposed:

H1b. When consumers do not have a shopping goal, they will (1) perceive the discount claims as more credible, (2) expect the average discount percentage to be higher, (3) judge the value of the product as higher, and (4) exhibit stronger intentions to purchase from the retailer when the processing of range-of-savings claims is disfluent (vs., fluent). These effects will occur regardless of consumers' regulatory focus.

4. Experiment

4.1. Design and participants

To test the hypotheses, an experiment was conducted using a 2 (Shopping Goal: Present vs. Absent) x 2 (Regulatory Focus: Prevention vs. Promotion) x 2 (Processing Fluency: Fluent vs. Disfluent) between-subjects design. A total of three hundred eighty-one South Korean adults were recruited through an online panel, and participants were randomly assigned to one of the experimental conditions. After excluding nine participants due to incomplete responses, the final sample size consisted of 372 participants (mean age = 39.9 years; female = 51.6 %).

4.2. Stimuli and procedure

The experiment was conducted online. Participants were first provided with a brief introduction and then immersed in a scenario where they encountered a 'range of savings' advertisement from an online clothing retailer while using their smartphones. Clothing was selected as the experimental product category due to its familiarity among participants and its frequent purchase through online shopping. The first independent variable, **shopping goal**, was manipulated by varying the context in which participants encountered the advertisement. Specifically, participants were either exposed to the advertisement while actively searching for information related to clothing purchases **(shopping goal present)** or while casually browsing social media **(shopping goal absent; see** Appendix A **for details).** The stimulus consisted of a full-screen smartphone display featuring a text-based background with a centrally located discount advertisement. The advertisement headline displayed the name of a fictional online clothing retailer and introduced the promotional event. The main body of the ad highlighted the range-of-savings claims (30–70 %), accompanied by a drawing of clothing to enhance realism. The drawing depicted either men's or women's clothing, tailored to match the participant's gender. To manipulate **processing fluency**, the font style of the discount information was varied to make the text either easy or difficult to process (see Appendix B for details). The third independent variable, **regulatory focus**, was assessed using a validated scale to measure individual tendencies (Haws et al., 2010). Based on their responses, participants were categorized into **promotion-focused** or **prevention-focused** groups using the median split method (promotion focus: Cronbach's $\alpha = .62$; prevention focus: Cronbach's $\alpha = .61$). After being exposed to the advertisement, participants completed a series of questionnaire items. Upon completing the experiment, they were thanked and dismissed.

4.3. Measures

Consumer responses to the discount advertisements were measured across several categories, following Biswas and Burton (1993). First, purchase intention toward the online retailer featured in the advertisement was assessed using a 7-point scale with four items ("I would click the link to visit [retailer]," "I am willing to purchase from [retailer]," "I am likely to shop at [retailer]," and "I would recommend [retailer] to others"; Cronbach's $\alpha = .94$). Second, the perceived value of the product was measured using a 7-point scale consisting of two items: "I would save a lot with this discount" and "The product offers high value" (Cronbach's $\alpha = .80$). Third, participants were asked to estimate the average discount percentage they expected to receive within the presented range of 30 %–70 %, in increments of 5 %. This measure captured the participants' expectations regarding the discount offered. Fourth, the credibility of the price discount was evaluated using a 7-point scale with two items: "I trust the discount claims" and "The discount claims seem credible" (Cronbach's $\alpha = .90$). To verify the manipulation of processing fluency, participants rated their experience of processing the advertisement on a 7-point semantic differential scale with three items: "clear to understand" vs. "difficult to understand," "fluent to process" vs. "disfluent to process," and "easy to comprehend" vs. "challenging to comprehend" (Cronbach's $\alpha = .90$; Mead and Hardesty, 2018).

4.4. Results

4.4.1. Manipulation checks

The manipulation of processing fluency in the discount advertisements was confirmed to be successful. Respondents perceived the discount ads as significantly more fluent in the fluently presented condition compared to the disfluent condition ($M_{\text{fluency}} = 3.37$, $SD = 1.29$ vs. $M_{\text{disfluency}} = 4.16$, $SD = 1.29$, $t\,(369) = -5.89$, $p < .001$).

4.4.2. Hypothesis testing

To test the hypotheses, a series of ANOVAs were performed with presence of a shopping goal, regulatory focus, and processing fluency of discount ads as the independent variables. Each of the four types of participant responses to the range-of-savings claims was used as a dependent variable in the analyses.

4.4.2.1. Purchase intention. A three-way interaction effect for purchase intention was found to be significant ($F\,(1, 364) = 4.71$, $p < .05$; $\eta_p^2 = .013$). To explore this interaction further, planned contrasts were conducted based on whether participants had a shopping goal. When a shopping goal was present, a significant interaction between regulatory focus and processing fluency was observed ($F\,(1, 364) = 8.36$, $p < .005$; $\eta_p^2 = .022$). Promotion-focused participants reported lower purchase intentions when the discount information was disfluently presented compared to fluently presented ($M_{\text{fluency}} = 4.05$, $SD = 1.15$ vs. $M_{\text{disfluency}} = 3.41$, $SD = 1.32$; $F\,(1, 364) = 4.41$, $p < .05$; $\eta_p^2 = .012$). In contrast, prevention-focused participants exhibited higher purchase intentions when the discount information was disfluently presented compared to fluently presented ($M_{\text{fluency}} = 3.48$, $SD = 1.76$ vs. $M_{\text{disfluency}} = 4.11$, SD

= 1.59; F (1, 364) = 3.97, $p < .05$; $\eta_p^2 = .011$). However, when participants were exposed to the discount advertisement without a shopping goal, the main effect of processing fluency was significant (F (1, 364) = 8.84, $p < .005$; $\eta_p^2 = .024$), while the two-way interaction effect between regulatory focus and processing fluency was not significant (F (1, 364) = .04, $p > .85$; $\eta_p^2 = .00$). Participants exhibited higher purchase intentions when the discount information was presented in a disfluent manner compared to when it was presented fluently, regardless of their regulatory focus ($M_{\text{fluency}} = 3.02$, $SD = 1.61$ vs. $M_{\text{disfluency}} = 3.66$, $SD = 1.39$; see Fig. 2).

4.4.2.2. Perceived product value. A significant three-way interaction was observed for perceived product value (F (1, 364) = 5.37, $p < .05$; $\eta_p^2 = .015$). To investigate this interaction further, contrast analyses were conducted based on whether participants had a shopping goal. When a shopping goal was present, a significant interaction between regulatory focus and processing fluency was found (F (1, 364) = 7.22, $p < .01$; $\eta_p^2 = .019$). Promotion-focused participants perceived the product as less valuable when the discount information was disfluently presented compared to fluently presented ($M_{\text{fluency}} = 4.38$, $SD = .94$ vs. $M_{\text{disfluency}} = 3.94$, $SD = 1.29$; F (1, 364) = 3.29, $p = .07$; $\eta_p^2 = .009$). In contrast, prevention-focused participants perceived the product as more valuable when the discount information was disfluently presented compared to fluently presented ($M_{\text{fluency}} = 3.87$, $SD = 1.26$ vs. $M_{\text{disfluency}} = 4.37$, $SD = 1.29$; F (1, 364) = 3.93, $p < .05$; $\eta_p^2 = .011$). When participants were exposed to the advertisement without a shopping goal, the main effect of processing fluency was significant (F (1, 364) = 6.17, $p < .05$; $\eta_p^2 = .017$), while the two-way interaction between regulatory focus and processing fluency was not significant (F (1, 364) = .36, $p > .50$; $\eta_p^2 = .001$). In this context, participants perceived the product as more valuable when the discount information was disfluently presented, regardless of their regulatory focus ($M_{\text{fluency}} = 3.80$, $SD = 1.15$ vs. $M_{\text{disfluency}} = 4.21$, $SD = 1.22$; see Fig. 3).

4.4.2.3. Expected average discount percentage. For the expected average discount percentage, a marginally significant three-way interaction was observed (F (1, 364) = 2.74, $p = .10$; $\eta_p^2 = .007$). Planned contrasts were conducted based on whether participants had a shopping goal. When participants had a shopping goal, the interaction between regulatory focus and processing fluency approached significance (F (1, 364) = 2.70, $p = .10$; $\eta_p^2 = .007$). Promotion-focused participants did not show a significant difference in their estimates of the average discount percentage based on fluency ($M_{\text{fluency}} = 46.84$ %, $SD = 11.26$ vs. $M_{\text{disfluency}} = 45.00$ %, $SD = 11.98$; F (1, 364) = .60, $p > .40$; $\eta_p^2 = .002$). However, prevention-focused participants expected to receive a slightly higher discount percentage when the information was presented disfluently compared to when it was presented fluently, although this difference did not reach statistical significance ($M_{\text{fluency}} = 44.44$ %, $SD = 12.85$ vs. $M_{\text{disfluency}} = 48.26$ %, $SD = 11.17$; F (1, 364) = 2.34, $p = .13$; $\eta_p^2 = .006$). When participants did not have a shopping goal, neither the main effect of fluency (F (1, 364) = .76, $p > .35$; $\eta_p^2 = .002$) nor the two-way interaction between regulatory focus and processing fluency was significant (F (1, 364) = .49, $p > .45$; $\eta_p^2 = .001$; see Fig. 4).

4.4.2.4. Credibility of discount. For the credibility of the discount claims, a significant three-way interaction was identified (F (1, 364) = 6.49, $p < .05$; $\eta_p^2 = .018$). Contrast analyses were conducted to examine this interaction based on whether participants had a shopping goal. When participants had a shopping goal, the interaction between regulatory focus and processing fluency was significant (F (1, 364) = 12.77, $p < .001$; $\eta_p^2 = .034$). Promotion-focused participants perceived the discount claims as less credible when the information was presented disfluently compared to fluently ($M_{\text{fluency}} = 4.12$, $SD = 1.08$ vs. $M_{\text{disfluency}} = 3.78$, $SD = 1.33$; F (1, 364) = 1.83, $p = .18$; $\eta_p^2 = .005$). In contrast, prevention-focused participants perceived the discount claims as more credible when the information was disfluently presented compared to fluently presented ($M_{\text{fluency}} = 3.51$, $SD = 1.44$ vs. $M_{\text{disfluency}} = 4.48$, $SD = 1.17$; F (1, 364) = 13.29, $p < .001$; $\eta_p^2 = .035$). When participants did not have a shopping goal, the main effect of fluency was significant (F (1, 364) = 6.14, $p < .05$; $\eta_p^2 = .017$), but the two-way interaction between regulatory focus and fluency was not significant (F (1, 364) = .00, $p > .95$; $\eta_p^2 = .00$). Participants perceived the discount claims as more credible when the information was disfluently presented, regardless of their regulatory focus ($M_{\text{fluency}} = 3.60$, $SD = 1.26$ vs. $M_{\text{disfluency}} = 4.04$, $SD = 1.18$; see Fig. 5).

4.4.2.5. Mediation test. According to the logic of the hypotheses, the independent variables will influence purchase intention through three mediators: the credibility of the discount claims, the expected average discount percentage, and the perceived value of the product. These mediators are expected to form a sequential mediation relationship, both in the presence and absence of a shopping goal. Although the effects of the independent variables on the average discount percentage were marginally significant or not significant, as reported in the previous section, some scholars suggest that non-significant relationships can still be included in the mediation model (Zhao et al., 2010).

To test this mediation relationship, analyses were conducted using PROCESS (model 6) with 5000 bootstrapped samples (Hayes, 2013).

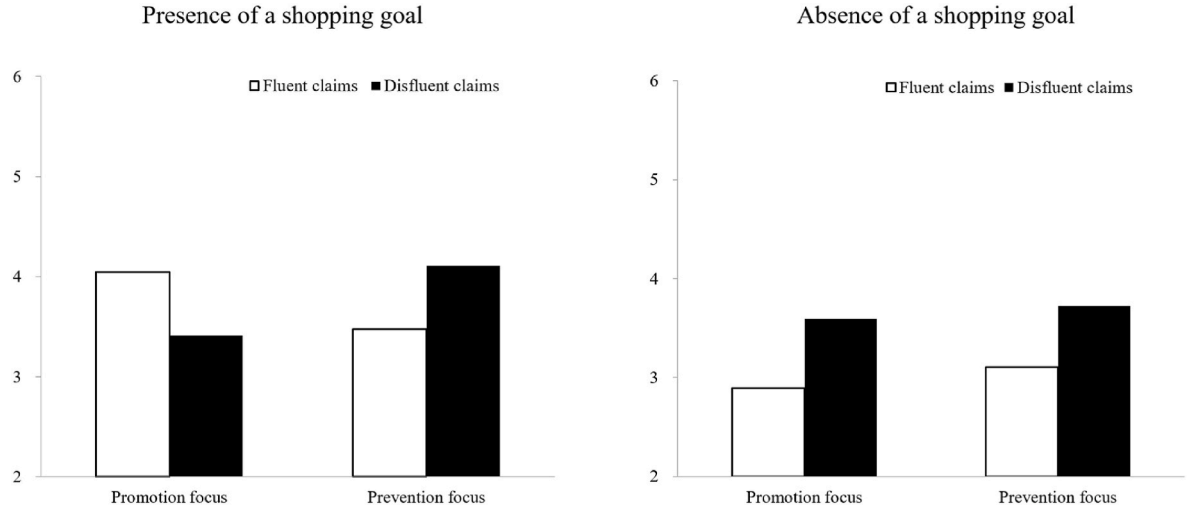


Fig. 2. The effect of regulatory focus, processing disfluency, and shopping goals on purchase intention.

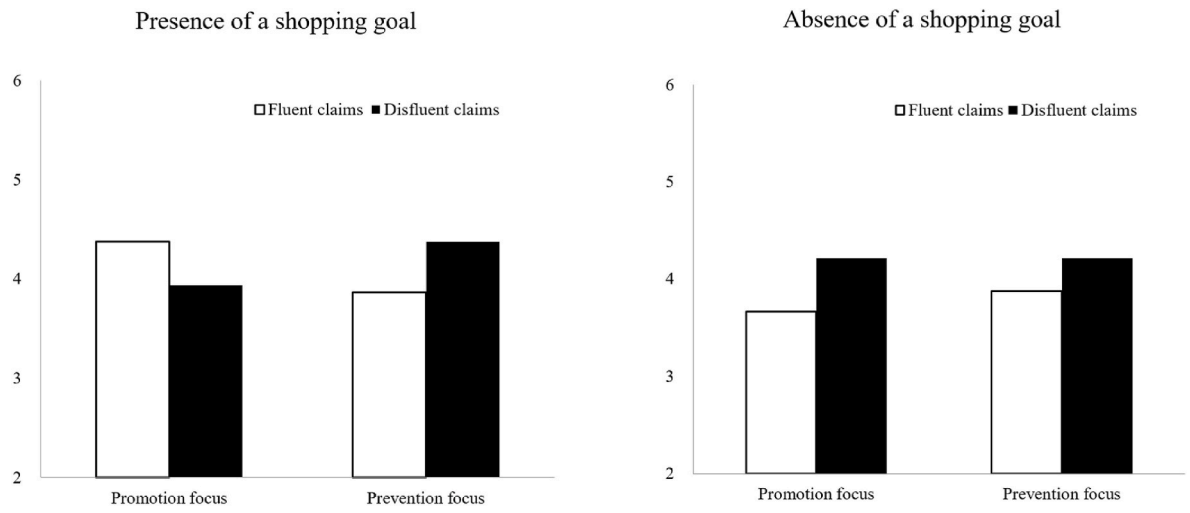


Fig. 3. The effect of regulatory focus, processing disfluency, and shopping goals on perceived product value.

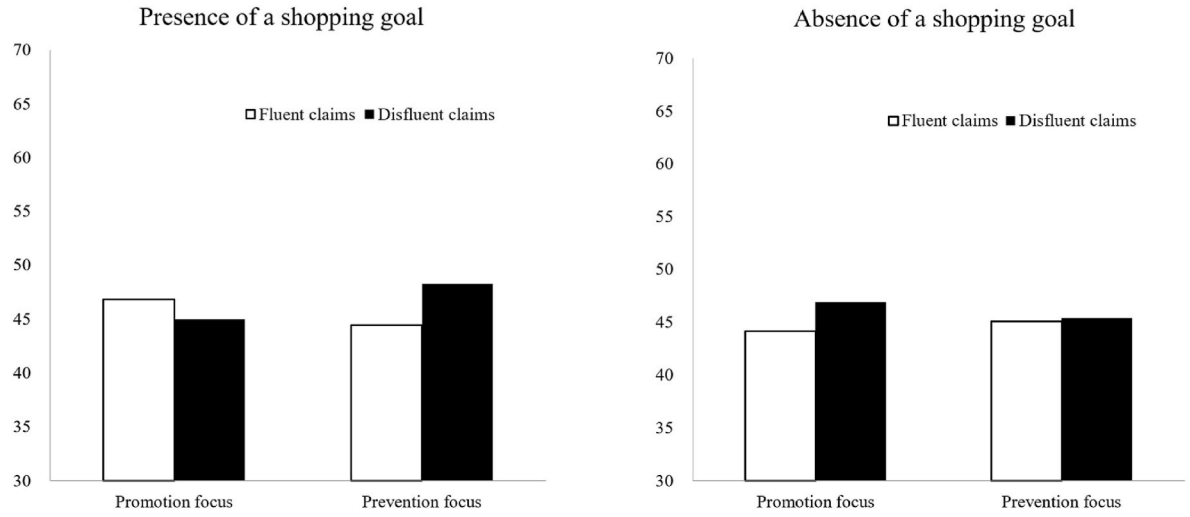


Fig. 4. The effect of regulatory focus, processing disfluency, and shopping goals on expected average discount percentage.

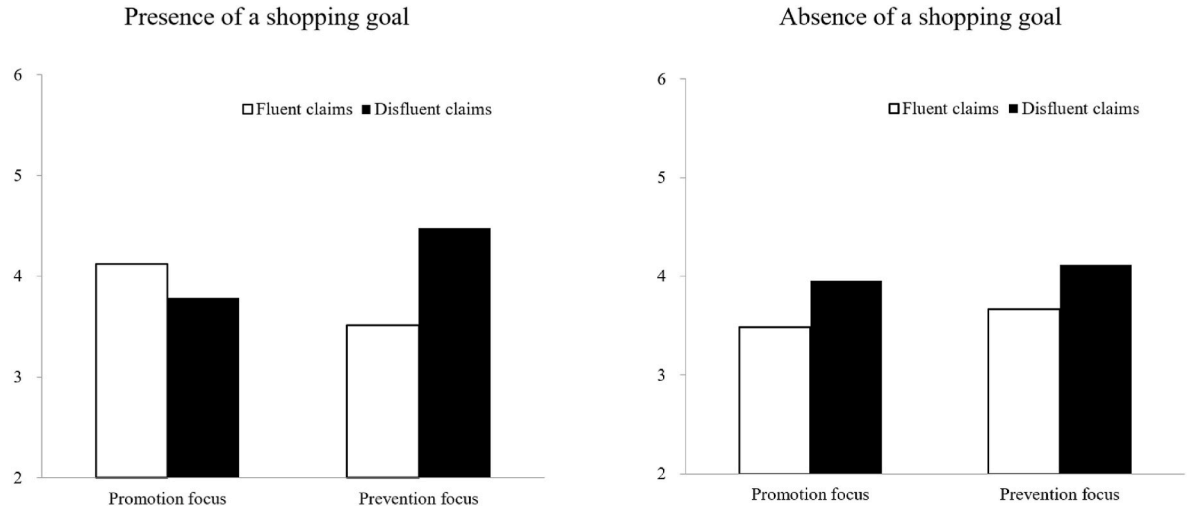


Fig. 5. The effect of regulatory focus, processing disfluency, and shopping goals on credibility of discount.

When a shopping goal was present, a multiple-step mediation analysis was performed, as the interaction between regulatory focus and processing fluency was significant for participants' purchase intentions. As expected, significant indirect effects were observed along the proposed mediation pathway for both prevention-focused participants (effect: 0.017, 90 % CI: 0.001 to 0.068) and promotion-focused participants (effect: 0.011, 90 % CI: 0.043 to −0.011). When a shopping goal was absent, where the main effect of processing fluency was significant irrespective of regulatory focus, the three mediators demonstrated a sequential relationship (effect: 0.006, 95 % CI: 0.001 to 0.023).

5. Conclusion

5.1. Summary of findings

This study demonstrated the complexity of consumer responses to range-of-savings claims in mobile advertisements. The effects of regulatory focus and processing fluency varied significantly depending on whether consumers had a shopping goal. First, when consumers with a shopping goal encountered savings claims that were processed fluently, promotion-focused consumers exhibited more positive responses than prevention-focused consumers. However, when the advertisements were processed disfluently, the positive responses of promotion-focused consumers diminished, while those of prevention-focused consumers improved. On the other hand, consumers without a shopping goal responded more positively to the savings claims when they were presented in a disfluent (vs. fluent) manner, regardless of their regulatory focus. These findings were consistent across three dependent variables, though the results for the average discount percentage expected were either marginally significant or non-significant. Furthermore, mediation analysis revealed that the (dis)fluency of savings claims influenced consumers' purchase intentions through a sequential mediation path involving the credibility of the savings claims, the average discount percentage expected, and the perceived value of the product. In conclusion, the findings largely supported H1a and H1b.

5.2. Theoretical and practical implications

The findings of this study offer several theoretical contributions to the literature on tensile price claims, regulatory focus, and processing fluency. First, while prior research has predominantly focused on comparing tensile price claims with precise discount claims or other types of discounts, as well as examining the effects of different subtypes of tensile price claims (Biswas and Burton, 1993, 1994; Dhar et al., 1999; Mobley et al., 1988), this study shifts the focus to the range-of-savings claims. Previous studies have explored moderating variables, but these were primarily related to the characteristics of the discount itself, such as discount levels or the proportion of discounted items, and store types. In contrast, this research identifies two critical consumer-related factors—regulatory focus and the presence of a shopping goal—and examines how these factors interact with the presentation style of discount information (i.e., processing fluency) while holding the discount content constant. This approach not only enriches the understanding of range-of-savings claims but also provides insights applicable to other range-based contexts, such as expected returns on financial products or range-based scales in surveys. Furthermore, while regulatory focus theory has been extensively studied in psychology and marketing since the 2000s, its application to the study of prices and discounts remains limited (Florack et al., 2013). By applying regulatory focus theory to tensile price claims, this research advances the understanding of its role in consumer behavior.

Second, this study contributes new insights into processing fluency, a topic of increasing research interest. Earlier studies suggested that consumers generally exhibit more favorable responses when information is fluently processed (Forster et al., 2013; Liu et al., 2024; Reber et al., 1998). However, subsequent research has identified contexts where disfluently processed information can yield positive effects (Oyserman, 2019; Pocheptsova et al., 2010; Song and Schwarz, 2009). Building on these findings, this study identifies the conditions under which disfluency can enhance consumer responses in the context of price discounts. Moreover, while three psychological mechanisms of processing fluency have been proposed, the conditions under which each mechanism operates remain unclear (Oppenheimer, 2008). Unlike previous research that often assumed specific mechanisms, this study systematically explores the mechanisms activated during the processing of price discounts. It finds that emotion-based mechanisms are not relevant in this context. Instead, two cognitive-based mechanisms—information processing style and theory-based reasoning—operate depending on the presence or absence of a shopping goal. Specifically, consumers with a shopping goal are affected by the information-processing style mechanism when exposed to disfluently presented discount information, whereas consumers without a shopping goal rely on the theory-based reasoning mechanism. These exploratory findings offer valuable new perspectives on the mechanisms underlying processing fluency.

Furthermore, we argued that this outcome was influenced by the different cognitive systems and information processing styles that consumers use depending on the presence or absence of a shopping goal (Celsi and Olson, 1988; Chaiken, 1980; Kahneman, 2003). Specifically, when there is a shopping goal, consumers tend to use a deliberate and systematic style, whereas when there is no shopping goal, they rely on automatic and heuristic methods. From a fit perspective, if disfluent information is assumed to require higher levels of information processing motivation and cognitive resources, it would align better with the condition where a shopping goal is present, but less so with the condition where no shopping goal exists (Hong and Lee, 2008). As a result, in the former condition, it would elicit favorable responses, while in the latter, it would generate negative responses. However, the results of this study suggest that the effects of processing fluency emerged in a much more complex manner than predicted by the fit theory.

Lastly, the findings also provide practical implications for retailers and policymakers. Retailers typically focus on stimulus-specific variables, such as discount levels and promotion durations, often assuming that discount information should be presented in a clear and easily understandable format. However, our results indicate that presenting discount information in a more difficult-to-process format can, under certain conditions, elicit more favorable consumer responses. Specifically, when targeting consumers who are incidentally exposed to savings ads without an active shopping goal, a disfluent format proves to be more effective. On the other hand, when targeting consumers with a shopping goal, ads should be personalized: savings claims should be presented fluently for promotion-focused consumers, whereas a disfluent format is better suited for prevention-focused consumers.

Marketers can leverage existing customer data to identify prospective consumers' regulatory focus. For instance, prevention-focused consumers tend to search for safety-oriented keywords such as "price," "durability," or "warranty," and they frequently visit comparison sites to evaluate various products. In contrast, promotion-focused consumers typically search for achievement-related keywords such as "quality," "performance," or "benefits" and generally have a shorter interval between search and purchase (Das, 2016; Xie and Kahle, 2014). Additionally, regulatory focus can be temporarily influenced by ad copy. For example, to activate a promotion-focused orientation, fluently processed savings claims should be paired with a gain-oriented message (e.g., "Take advantage of this opportunity! 30–70 % savings"). In contrast, when the ad environment induces disfluent processing, a loss-averse message should be used instead (e.g., "Don't miss this opportunity! 30–70 % savings").

A similar approach can also be applied when offering coupons. For instance, when targeting consumers with longer shopping durations (indicative of a shopping goal), it may be effective to design coupons that encourage fluent processing for promotion-focused consumers

while using a disfluent design for prevention-focused consumers. These findings also extend to other range-based contexts. For example, when presenting expected returns on financial products as a range, firms and policymakers should consider how consumer perceptions of average returns may vary depending on regulatory focus or processing fluency. These considerations could inform product development, communication strategies, and the formulation of government policies. Additionally, in surveys that use range-based scales to measure the likelihood of certain events, promotion-focused respondents may overestimate probabilities, whereas prevention-focused respondents may underestimate them. Accounting for these tendencies when analyzing survey data could improve the accuracy of interpreting results.

5.3. Limitations and directions for future research

This study has several limitations that should be considered. First, the manipulation of processing disfluency in this study was achieved by varying the font style used to present discount information in advertisements. However, other factors, such as reversing the order of the minimum and maximum numbers in the savings range or altering the font size of these numbers, could also influence processing fluency (Biswas et al., 2013; Mueller et al., 2014). While there is limited evidence to suggest that these variations would substantially alter the findings, future research should investigate these manipulations to enhance the generalizability of the results. In particular, combining disparate elements of processing fluency, such as font style and numerical order, could produce complex interaction effects that warrant further exploration. Conducting field experiments or employing alternative methodologies could also provide valuable insights into how various consumer-related factors influence responses, especially given advancements in online data analysis that allow for estimating consumers' regulatory focus tendencies and the presence of shopping goals (Fiestas and Tuzovic, 2021).

Second, the average discount percentage expected, one of the dependent variables examined in this study, yielded marginally significant or non-significant results for three-way interactions and contrasts based on the presence of a shopping goal. Two potential explanations may account for this outcome. First, the midpoint (50 %) of the discount range used in the experiment (30 %–70 %) might have served as an easily calculable anchor, leading to similar adjustments across experimental conditions. Second, Asian participants, who are generally recognized for their strong numerical computation skills, may have relied on precise calculations when estimating the discount percentage, making them less sensitive to the experimental manipulations (Geary et al., 1999). Furthermore, although this study categorized participants into promotion-focused and prevention-focused groups based on their regulatory focus dispositions, it is worth noting that Asians, in general, tend to be more prevention-focused (Lee et al., 2000). These tendencies may lead them to expect a slightly lower average discount percentage and be less influenced by the independent variables. Additionally, the effect of regulatory focus was weak when consumers without shopping goals were exposed to discount advertisements. This may be due to their lack of motivation or cognitive resources for information processing, leaving little room for regulatory focus—a motivational system—to influence their responses (Roskes et al., 2012). These findings highlight the need for future research to clarify these unexpected results.

Finally, this study assumed that consumers' processing of discount information is primarily a cognitive process, thereby excluding emotion-based mechanisms of processing disfluency. However, prior research suggests that regulatory focus can affect consumers' reliance on emotion-versus cognition-based processing tendencies (Greifeneder and Keller, 2012). Future research could take a broader perspective to explore the interplay between regulatory focus and processing fluency mechanisms. In this regard, cognitive load theory may offer valuable insights (Sweller, 2011). How might the findings of this study change if consumers encounter disfluent claims while their cognitive resources are depleted? In the presence of a shopping goal, resource depletion may disrupt consumers' deliberate and systematic processing, potentially preventing them from engaging in the adjustment process and leading them to maintain their initial judgment. Alternatively, consumers may rely on an emotion-based mechanism to process disfluency. In contrast, when consumers lack a shopping goal—typically associated with automatic and effortless processing—resource depletion may have little to no impact. Furthermore, since prevention-focused consumers are more sensitive to cognitive resource availability than promotion-focused consumers, the effects of resource depletion may vary depending on regulatory focus (Roskes et al., 2012).

Moreover, it would be valuable to explore whether factors identified in prior research on tensile price claims could moderate the effects of range-of-savings claims. For example, the proportion of discounted items within the entire inventory might interact with processing disfluency or regulatory focus. Disfluency could encourage more deliberate and systematic information processing, amplifying or mitigating the effects of discount proportions. Additionally, promotion-focused consumers might respond more positively to low (vs. high) discount proportions due to perceived scarcity, while prevention-focused consumers could view higher proportions of discounted items as safer, thus reducing negative reactions to range-of-savings claims. On the other hand, the factors examined in this study could also contribute to understanding consumer responses to tensile price claims compared to precise price claims. For instance, prevention-focused consumers might evaluate precise discount claims or minimum savings claims more favorably than promotion-focused consumers. Furthermore, consumers with a shopping goal, whose implementation mindset is activated, might exhibit more positive responses to precise discount claims compared to tensile price claims due to the higher specificity of the former claims (Dhar et al., 2007).

CRediT authorship contribution statement

Kikyoung Park: Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Gangseog Ryu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist in translating portions of the manuscript written in Korean into English and proofreading them. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing interest

We confirm that this manuscript is original and has not been submitted or published elsewhere. We also declare that there are no conflicts of interest related to this work. All co-authors have reviewed and approved the manuscript, and we have adhered to ethical guidelines in conducting this research.

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Appendix A. *Scenarios Used to Manipulate the Presence or Absence of a Shopping Goal*

<Presence of a Shopping Goal>

You are using your smartphone to search for fashion trends and popular items, browsing fashion-related articles with the intention of purchasing clothes. While doing this, you happen to notice an advertisement for 'OO Shopping Mall' on part of the screen. 'OO Shopping Mall' is a fairly well-known, fashion-focused online shopping platform.

<Absence of a Shopping Goal>

You have some free time, so you are checking your social media and emails on your smartphone, as well as reading articles that interest you. While doing this, you happen to notice an advertisement for 'OO Shopping Mall' on part of the screen. 'OO Shopping Mall' is a fairly well-known, fashion-focused online shopping platform.

Appendix B. *Stimuli for Experiment*

<Fluent ad claims: Original version>



<Disfluent ad claims: Original version>



<Fluent ad claims: Translated version>



<Disfluent ad claims: Translated version>



Data availability

The data that has been used is confidential.

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